

$$Q_1 - \frac{1}{r} \frac{d}{dr} (k_f r_f \frac{dT_f}{dr}) + \dot{q} = 0 \Rightarrow \int \frac{d}{dr} \left(r_f \frac{dT_f}{dr} \right) = - \frac{\dot{q} r_f}{k_f}$$

$$\Rightarrow r_f \frac{dT_f}{dr} = - \frac{\dot{q} r_f^2}{2k_f} + C_1 \Rightarrow \int \frac{dT_f}{dr} = \int - \frac{\dot{q} r_f}{2k_f} + \int \frac{C_1}{r_f}$$

$$\Rightarrow T_f(r) = - \frac{\dot{q} r_f^2}{4k_f} + C_1 \ln(r_f) + C_2$$

+ Heat Egn:

$$\frac{1}{r} \frac{\partial}{\partial r} (k_c r_c \frac{\partial T_c}{\partial r}) = 0 \quad (\text{Steady State})$$

$$\Rightarrow \int \frac{\partial}{\partial r} (k_c r_c \frac{\partial T_c}{\partial r}) = 0 \Rightarrow k_c r_c \frac{\partial T_c}{\partial r} = C_3 \Rightarrow \int \frac{\partial T_c}{\partial r} = \int \frac{C_3}{k_c r_c}$$

$$\Rightarrow T_c(r) = \frac{C_3}{k_c} \ln(r_c) + C_4$$

28

• BC₁: at $r_f = 0$, $\frac{dT_f}{dr} = 0 \Rightarrow C_1 = 0$

• BC₂: $-k_c \frac{dT_c}{dr} \Big|_{r=r_c} = h(T_c - T_\infty)$

$$\Rightarrow -\cancel{k_c} \left(\frac{C_3}{\cancel{k_c} r_c} \right) = h \left(\frac{C_3}{k_c} \ln(r_c) + C_4 - T_\infty \right)$$

$$\Rightarrow - \frac{C_3}{r_c} = \frac{h C_3}{k_c} \ln(r_c) + h C_4 - h T_\infty$$

$$BC_3: \text{ at } r_c = r_f \quad T_c(r) = T_f(r)$$

$$\frac{c_3}{k_c} \ln(r_f) + c_4 = - \frac{\dot{q} r_f^2}{4k_f} + c_2$$

$$BC_4: \text{ at } r_c = r_f \quad k_c \frac{dT_c}{dr} = k_f \frac{dT_f}{dr}$$

$$k_c \left(\frac{c_3}{k_c r_c} \right) = k_f \left(- \frac{\dot{q} r_f}{2k_f} \right) \Rightarrow c_3 = - \frac{\dot{q} r_f^2}{2}$$

$$\Rightarrow \frac{\dot{q} r_f^2}{2r_c} = \frac{h}{k_c} \left(- \frac{\dot{q} r_f^2}{2} \right) \ln(r_c) + h c_4 - h T_\infty$$

$$\Rightarrow c_4 = \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln(r_c) + T_\infty$$

$$\frac{c_3}{k_c} \ln r_f + c_4 = - \dot{q} \frac{r_f^2}{4k_f} + c_2 \Rightarrow - \frac{\dot{q} r_f^2}{2k_c} \ln(r_f) + \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln(r_c) + T_\infty = - \dot{q} \frac{r_f^2}{4k_f} + c_2$$

$$\Rightarrow c_2 = \frac{\dot{q} r_f^2}{2k_f} \ln\left(\frac{r_c}{r_f}\right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty + \frac{\dot{q} r_f^2}{4k_f}$$

$$\Rightarrow T_f(r) = - \frac{\dot{q} r_c^2}{4k_f} + c_1 \ln(r_f) + c_2 \Rightarrow T_f(r) = - \frac{\dot{q} r_c^2}{4k_f} + \frac{\dot{q} r_f^2}{2k_c} \ln\left(\frac{r_c}{r_f}\right) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty + \frac{\dot{q} r_f^2}{4k_f}$$

$$\therefore T_f(r) = \frac{\dot{q} r_f^2}{2k_c} \ln\left(\frac{r_c}{r_f}\right) + \dot{q} (r_f^2 - r^2) + \frac{\dot{q} r_f^2}{2r_c h} + T_\infty$$

$$\rightarrow T_c(r) = \frac{c_3}{k_c} \ln(r_f) + c_4 = - \frac{\dot{q} r_f^2}{2k_c} \ln(r_f) + \frac{\dot{q} r_f^2}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln(r_c) + T_\infty$$

$$\Rightarrow T_c(r) = \frac{\dot{q} r_f}{2r_c h} + \frac{\dot{q} r_f^2}{2k_c} \ln\left(\frac{r_c}{r_f}\right) + T_\infty$$



b) at $r=0$ $T=T_{max}$

$$T_f(r_{max}) = \frac{\dot{q} r_f^2}{2k_c} \ln\left(\frac{r_c}{r_f}\right) + \frac{\dot{q}}{4k_f} (r_f^2) + \frac{\dot{q} r_f^2}{2r_c R} + T_{\infty}$$

$$T_f(r_{max}) = \frac{(2 \times 10^8)(0.006)^2}{(2)(25)} \ln\left(\frac{0.009}{0.006}\right) + \frac{(2 \times 10^8)(0.006)^2}{(4)(3)} + \frac{(2 \times 10^8)(0.006)^2}{(2)(0.009)(2000)} + 30$$

$$T_f(r_{max}) = 1158.4 \text{ K}$$

Q_z - In - out + Gen - Cons = $\Delta \dot{Q}$

$$\dot{q} - \left(\dot{q} + \frac{d\dot{q}}{dx} dx\right) + \dot{q} - \dot{q}'' = 0$$

$$\dot{q} - \dot{q} - \frac{d\dot{q}}{dx} dx + \dot{q} A dx - \dot{q}'' (\delta A) dx = 0$$

$$\frac{1}{dx} \left[\frac{d}{dx} \left(kA \frac{dT}{dx} \right) dx + \dot{q} A dx - \dot{q}'' (\delta A) dx \right] = 0 \Rightarrow \frac{d}{dx} kA \frac{dT}{dx} + \dot{q} A - \dot{q}'' (\delta A) = 0$$

$$\frac{d}{dx} kA \frac{dT}{dx} = \dot{q}'' (\delta A) - \dot{q} A \Rightarrow \frac{d}{dx} k\pi \left[r \left(1 - \frac{x}{L} \right) \right]^2 \frac{dT}{dx} = \dot{q}'' \left[2\pi r \left(1 - \frac{x}{L} \right) \right] - \dot{q} \pi \left[r \left(1 - \frac{x}{L} \right) \right]^2$$

$$\Rightarrow \frac{d}{dx} k\pi r^2 (1-x)^2 \frac{dT}{dx} = \dot{q}'' 2\pi r (1-x) - \dot{q} \pi r^2 (1-x)^2$$

$$\frac{k\pi r^2 (1-x)^2 \frac{dT}{dx}}{k\pi r^2 (1-x)^2} = \frac{-\dot{q}'' \frac{2\pi r}{2} (1-x)^2 + \dot{q} \frac{\pi r^2}{3} (1-x)^3 + C_1}{k\pi r^2 (1-x)^2}$$

$$\frac{dT}{dx} = -\frac{\dot{q}''}{kr} + \frac{\dot{q}}{3k} (1-x) + \frac{C_1}{k\pi r^2 (1-x)^2}$$

$$T(x) = -\frac{\dot{q}'' x}{kr} - \frac{\dot{q}}{6k} (1-x)^2 + \frac{C_1}{k\pi r^2 (1-x)^2} ; \quad \left. \frac{dT}{dx} \right|_{x=1} = 0 \rightarrow -\frac{\dot{q}''}{kr} + 0 + \frac{C_1}{k\pi r^2 (1-x)^2} = 0$$

$$\Rightarrow C_1 = 0 \rightarrow T(x=0) = 30$$

$$T(0) = -\frac{q''}{kx} - \frac{\dot{q}}{6x} (1-x)^2 + C_2 \Rightarrow 30 = -\frac{\dot{q}}{6k} + C_2 \rightarrow C_2 = 30 + \frac{\dot{q}}{6k}$$

$$T(x) = -\frac{q''}{kx} - \frac{\dot{q}}{6k} (1-x)^2 + \frac{\dot{q}}{6k} + 30$$

$$T(x) = -\frac{(0.1)(1)}{(20)(0.5)} - \frac{(100)}{6(20)} (1-1)^2 + \frac{(100)}{6(20)} + 30 = 30.8^\circ\text{C}$$

$$Q_3 - \dot{E}_{\text{out}} = \dot{E}_{\text{ss}} \Rightarrow -hA_s (T - T_\infty) = \rho V_c \frac{dT}{dt}$$

$$\Rightarrow \frac{\rho V_c}{hA_s} \frac{d\theta}{dt} = -\theta \quad - T_\infty \text{ is constant}$$

$$\Rightarrow \frac{d\theta}{dt} = \frac{dT}{dt} \Rightarrow -\int_0^t dt = \frac{\rho V_c}{hA_s} \int_{\theta_i}^{\theta} \frac{d\theta}{\theta}$$

$$\Rightarrow -t \Big|_0^t = \frac{\rho V_c}{hA_s} \left[\ln \theta \right]_{\theta_i}^{\theta}$$

$$\Rightarrow -t = \frac{\rho V_c}{hA_s} (\ln \theta - \ln \theta_i) \Rightarrow t = \frac{\rho V_c}{hA_s} \cdot \ln \left(\frac{\theta_i}{\theta} \right)$$

$$\Rightarrow t = \frac{\rho V_c}{hA_s} \ln \left(\frac{\theta_i}{\theta} \right)$$

$$\Rightarrow \frac{1}{A_s} = \frac{t \rho}{\rho V_c \ln \left(\frac{\theta_i}{\theta} \right)}$$

25

$$R = \frac{1}{\left(\frac{1}{R_o}\right) + \left(\frac{1}{R_i}\right)} = \frac{1}{\left(\frac{1}{2000}\right) + \left(\frac{1}{10000}\right)} = 1666.7 \text{ W/m}^2 \cdot \text{K}$$

$$\Rightarrow \frac{1}{A_s} = \frac{(3600 \text{ s})(1666.7 \text{ W/m}^2 \cdot \text{K})}{(1200 \text{ kg/m}^2)(2.25 \text{ m}^3)(2200 \text{ J/kg}) \ln\left(\frac{200}{50}\right)} \Rightarrow A_s = 1.37 \text{ m}^2$$

$$A_s = \pi D L \Rightarrow 1.37 = \pi D L \Rightarrow L = 21.8 \text{ m}$$